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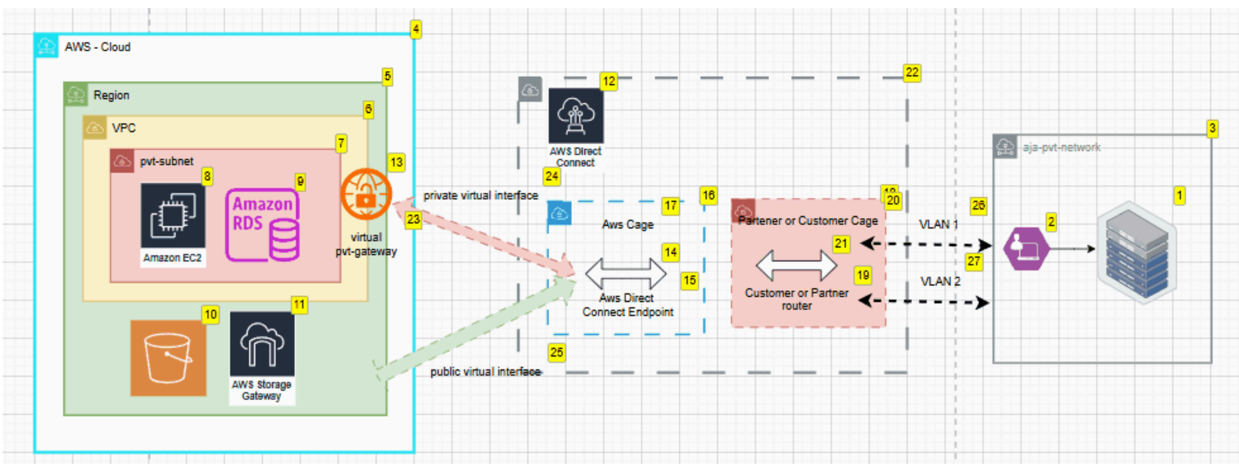
## 🧠 AWS Direct Connect – Solutions Architect Level Overview

### ✅ What is AWS Direct Connect?

**AWS Direct Connect** (DX) is a **dedicated network connection** between your **on-premises data center (or co-location)** and AWS.

It provides a **private, low-latency, high-throughput** alternative to the public internet when accessing AWS services.

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### 🎯 Why Use AWS Direct Connect?

| Benefit | Description |
|---|---|
| **Low Latency** | Reduces latency compared to the public internet |
| **High Bandwidth** | Supports up to 100 Gbps connections (dedicated) |
| **Secure** | Traffic doesn't traverse the public internet |
| **Consistent Performance** | Predictable throughput and minimal jitter |

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| **Hybrid Cloud** | Ideal for hybrid environments requiring private connectivity |

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## 📌 Key Concepts

### 1. Connection Types

| Type | Speed | Description |
|---|---|---|
| **Dedicated Connection** | 1 Gbps, 10 Gbps, or 100 Gbps | Provisioned through AWS Management Console |
| **Hosted Connection** | 50 Mbps – 10 Gbps | Provisioned via AWS Partner (Direct Connect Partner) |
| **Hosted Virtual Interface (VIF)** | 50 Mbps – 5 Gbps | Shared with multiple customers by DX partners |

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### 2. Virtual Interfaces (VIFs)

AWS Direct Connect uses **Virtual Interfaces (VIFs)** to route traffic:

| VIF Type | Use Case | Connects to |
|---|---|---|
| **Private VIF** | Accessing VPC via private IP | VPC via VGW or TGW |
| **Public VIF** | Accessing AWS public services (e.g., S3, DynamoDB) | AWS Public Services over public IP |
| **Transit VIF** | Multiple VPCs via Transit Gateway | TGW attachment for large-scale networks |

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### 3. Router Requirements

- Border Gateway Protocol (BGP) is required
- ASN (Autonomous System Number) must be configured
- Redundancy and failover need to be designed (DX + VPN fallback)

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### 🛠️ Common Architectures

#### 📘 Scenario 1: Private VIF to a Single VPC

- On-prem → Direct Connect → AWS Router → VGW → VPC
- Use case: Latency-sensitive workloads accessing EC2 or RDS privately.

#### 📘 Scenario 2: Transit VIF for Multi-VPC Architecture

- On-prem → DX → TGW → Multiple VPCs
- Use case: Centralized network control, large-scale hybrid connectivity.

#### 📘 Scenario 3: Redundant Direct Connect

- Two DX locations in different AWS regions or availability zones
- Use VPN over internet as backup if DX fails (recommended by AWS)

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### 📎 Integration with AWS Services

| Service | Integration |
| --- | --- |

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|---|---|
| **Amazon VPC** | Direct access to EC2, RDS, etc. via private VIF |
| **Amazon S3, DynamoDB** | Accessed using public VIF |
| **Transit Gateway** | Aggregate access to multiple VPCs using transit VIF |
| **Route 53** | Works with private hosted zones |
| **CloudWatch** | Monitor DX performance via metrics and alarms |

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### 🧱 Key Components

- **Customer Router** – Your on-premises router
- **AWS Router** – AWS endpoint you peer with
- **DX Location** – Physical facility where AWS provides Direct Connect access
- **LOA-CFA (Letter of Authorization – Connecting Facility Assignment)** – Required to provision a circuit to AWS

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### 🧪 DX Best Practices (Architect Perspective)

- Design for **high availability** (DX + VPN fallback)
- Monitor via **CloudWatch & DX metrics**
- Use **Transit Gateway** for scalable multi-VPC routing

- Consider **encryption overlays** (DX is not encrypted by default)
- Use **BGP attributes** (like AS-PATH prepending, MED) for route control

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### 🧠 Sample Exam Tips / Notes

- DX does **not** encrypt traffic – use IPsec if needed
- Always recommend **redundancy** – DX + VPN
- Know the **difference between VIF types**
- Use **Transit VIF** only with **Transit Gateway**
- Hosted connections are provisioned by **Direct Connect Partners**, not AWS directly
- Public VIFs can access AWS services globally (like S3)

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### 🧾 Summary Sheet for Notes

| Concept | Details |
|---|---|
| VIF Types | Private, Public, Transit |
| DX Speeds | 50 Mbps – 100 Gbps |
| Encryption | Not by default |
| BGP | Required for routing |

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|---|---|
| Redundancy | DX + VPN recommended |
| TGW Support | Yes, via Transit VIF |
| Monitoring | CloudWatch supported |